

BitsB2B Marketplace

PostgreSQL Relational Database Schema Specification

Section 1: Authentication, Credentials & RBAC Tables

Table: 1. users

Primary user identity record (Buyers, Sellers, Dual Traders & Admins).

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY DEFAULT gen_random_uuid()	Unique user identifier
full_name	VARCHAR(150)	NOT NULL	Full name of account owner
phone	VARCHAR(30)	NOT NULL UNIQUE	Mobile phone number (SMS OTP login)
email	VARCHAR(150)	NULLABLE UNIQUE	Account email address
avatar_url	TEXT	NULLABLE	Profile avatar image link
is_active	BOOLEAN	DEFAULT TRUE	Account active status flag
is_verified	BOOLEAN	DEFAULT FALSE	Identity verification status flag
created_at	TIMESTAMPTZ	DEFAULT CURRENT_TIMESTAMP	Account creation timestamp
updated_at	TIMESTAMPTZ	DEFAULT CURRENT_TIMESTAMP	Last profile update timestamp

Table: 2. credentials

Stores passwords, hashes, and OTP secrets linked to user identities.

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY DEFAULT gen_random_uuid()	Unique credential record ID
user_id	UUID	NOT NULL REFERENCES users(id) ON DELETE CASCADE	Foreign Key to users table
password_hash	VARCHAR(255)	NULLABLE	Salted password hash (Argon2/BCrypt)
password_salt	VARCHAR(255)	NULLABLE	Cryptographic salt string
otp_secret	VARCHAR(100)	NULLABLE	SMS OTP secret token
failed_attempts	INTEGER	DEFAULT 0	Consecutive failed login counter
created_at	TIMESTAMPTZ	DEFAULT CURRENT_TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMPTZ	DEFAULT CURRENT_TIMESTAMP	Last credentials update timestamp

Table: 3. authentications

Tracks active user login sessions, JWT refresh tokens, and device metadata.

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY DEFAULT gen_random_uuid()	Session identifier
user_id	UUID	NOT NULL REFERENCES users(id) ON DELETE CASCADE	Foreign Key to users table
refresh_token_hash	VARCHAR(255)	NOT NULL UNIQUE	Hashed JWT refresh token
device_info	VARCHAR(255)	NULLABLE	User Agent device metadata
ip_address	VARCHAR(45)	NULLABLE	IPv4 / IPv6 client address
expires_at	TIMESTAMPTZ	NOT NULL	Session expiration timestamp
revoked_at	TIMESTAMPTZ	NULLABLE	Token revocation timestamp
created_at	TIMESTAMPTZ	DEFAULT CURRENT_TIMESTAMP	Session start timestamp

Table: 4. roles

Defines system and domain roles (ROLE_ADMIN, ROLE_BUYER, ROLE_SELLER, ROLE_DUAL_TRADER).

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY DEFAULT gen_random_uuid()	Role ID
name	VARCHAR(50)	NOT NULL UNIQUE	Display role title
description	TEXT	NULLABLE	Role scope description
created_at	TIMESTAMPTZ	DEFAULT CURRENT_TIMESTAMP	Creation timestamp

Table: 5. permissions

Granular system access permissions across catalog, orders, escrow & admin module.

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY DEFAULT gen_random_uuid()	Permission ID
name	VARCHAR(100)	NOT NULL UNIQUE	Display permission title
code	VARCHAR(100)	NOT NULL UNIQUE	Permission code (e.g. order:release_escrow)
module	VARCHAR(50)	NOT NULL	Module category (catalog, escrow, admin)
description	TEXT	NULLABLE	Permission behavior description
created_at	TIMESTAMPTZ	DEFAULT CURRENT_TIMESTAMP	Creation timestamp

Table: 6. role_permissions

Junction Table mapping granular permissions to roles (Many-to-Many).

Column Name	Data Type	Constraints & Keys	Description
role_id	UUID	NOT NULL REFERENCES roles(id) ON DELETE CASCADE	Foreign Key to roles table
permission_id	UUID	NOT NULL REFERENCES permissions(id) ON DELETE CASCADE	Foreign Key to permissions table
granted_at	TIMESTAMPTZ	DEFAULT CURRENT_TIMESTAMP	Grant timestamp
rolePermit_id	-	PRIMARY KEY (role_id, permission_id)	Unique mapping constraint

Table: 7. user_roles

Junction Table mapping roles to users (Many-to-Many, supports Admin & Dual Trading).

Column Name	Data Type	Constraints & Keys	Description
user_id	UUID	NOT NULL REFERENCES users(id) ON DELETE CASCADE	Foreign Key to users table
role_id	UUID	NOT NULL REFERENCES roles(id) ON DELETE CASCADE	Foreign Key to roles table
assigned_by	UUID	NULLABLE REFERENCES users(id)	Foreign Key to Admin users ID
assigned_at	TIMESTAMPTZ	DEFAULT CURRENT_TIMESTAMP	Assignment timestamp
userRole_id	-	PRIMARY KEY (user_id, role_id)	Unique mapping constraint

Section 2: Enterprise Business & Marketplace Core Tables

Table: 8. businesses

Enterprise profile with dual-role capability flags & TIN verification status.

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY DEFAULT gen_random_uuid()	Business profile ID
user_id	UUID	NOT NULL REFERENCES users(id) ON DELETE CASCADE	Foreign Key to owner users ID
name	VARCHAR(200)	NOT NULL	Legal enterprise name
primary_role	VARCHAR(50)	NOT NULL	Primary business classification
capabilities	TEXT[]	DEFAULT ARRAY['can_buy', 'can_sell']	Capabilities array
can_buy	BOOLEAN	DEFAULT TRUE	Purchasing capability flag
can_sell	BOOLEAN	DEFAULT TRUE	Selling capability flag
phone	VARCHAR(30)	NOT NULL	Business contact phone
region	VARCHAR(100)	NOT NULL	Ethiopian region
city	VARCHAR(100)	NOT NULL	City name
subcity	VARCHAR(100)	NULLABLE	Subcity jurisdiction
kebele	VARCHAR(50)	NULLABLE	Kebele administrative unit
landmark	VARCHAR(200)	NULLABLE	Nearby landmark
tin_number	VARCHAR(50)	NULLABLE	Tax Identification Number
trade_license_number	VARCHAR(100)	NULLABLE	Business Registration License
is_verified	BOOLEAN	DEFAULT FALSE	Business TIN verification flag
verification_status	VARCHAR(30)	DEFAULT 'pending'	Status (verified, pending, rejected)
verified_at	TIMESTAMPTZ	NULLABLE	Admin approval timestamp
verified_by	UUID	NULLABLE REFERENCES users(id)	Foreign Key to Admin users ID
established_year	INTEGER	NULLABLE	Year established
average_response_time	VARCHAR(50)	DEFAULT '< 2 hours'	Response SLA

Column Name	Data Type	Constraints & Keys	Description
response_rate	VARCHAR(20)	DEFAULT '100%'	Response rate percentage
rating	NUMERIC(3, 2)	DEFAULT 5.00	Supplier rating score
total_orders_completed	INTEGER	DEFAULT 0	Fulfillment order tally
description	TEXT	NULLABLE	Company bio / capacity
created_at	TIMESTAMPTZ	DEFAULT CURRENT_TIMESTAMP	Profile creation timestamp

Table: 9. categories

Wholesale product categories.

Column Name	Data Type	Constraints & Keys	Description
id	VARCHAR(50)	PRIMARY KEY	Category ID (e.g. cat-industrial)
name	VARCHAR(150)	NOT NULL	Category display title
description	TEXT	NULLABLE	Category scope summary

Table: 10. products

B2B wholesale product listings published by enterprises with can_sell = TRUE.

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY DEFAULT gen_random_uuid()	Product ID
seller_business_id	UUID	NOT NULL REFERENCES businesses(id) ON DELETE CASCADE	Foreign Key to seller businesses ID
seller_user_id	UUID	NOT NULL REFERENCES users(id) ON DELETE CASCADE	Foreign Key to seller users ID
category_id	VARCHAR(50)	NOT NULL REFERENCES categories(id) ON DELETE RESTRICT	Foreign Key to categories table
name	VARCHAR(255)	NOT NULL	Wholesale item title
description	TEXT	NOT NULL	Product description
price	NUMERIC(12, 2)	NOT NULL	Base unit price
currency	VARCHAR(10)	DEFAULT 'ETB'	Currency code
moq	INTEGER	NOT NULL DEFAULT 1	Minimum Order Quantity
unit	VARCHAR(50)	NOT NULL	Quantity unit (e.g. pieces)
stock_status	VARCHAR(30)	DEFAULT 'in_stock'	Status (in_stock, low_stock, out_of_stock)
stock_quantity	INTEGER	DEFAULT 0	Quantity in stock
stock_last_updated	VARCHAR(50)	DEFAULT 'Just now'	Human-readable update timestamp
lead_time	VARCHAR(100)	DEFAULT '3-5 business days'	Delivery lead time
delivery_zones	JSONB	NULL	Array of supported regions
images	JSONB	NOT NULL	Array of image URLs
specifications	JSONB	NOT NULL	Key-value specifications JSON
created_at	TIMESTAMPTZ	DEFAULT CURRENT_TIMESTAMP	Publication timestamp

Table: 11. product_price_tiers

Dynamic volume discount tier pricing per product.

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY DEFAULT gen_random_uuid()	Tier record ID
product_id	UUID	NOT NULL REFERENCES products(id) ON DELETE CASCADE	Foreign Key to products table
min_quantity	INTEGER	NOT NULL	Minimum threshold quantity
price_per_unit	NUMERIC(12, 2)	NOT NULL	Discounted unit price

Table: 12. orders

Escrow protected purchase orders placed on the platform.

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY DEFAULT gen_random_uuid()	Order ID
order_number	VARCHAR(50)	NOT NULL UNIQUE	Human order code (e.g. ORD-84920)
buyer_user_id	UUID	NOT NULL REFERENCES users(id) ON DELETE	Foreign Key to buyer users ID

Column Name	Data Type	Constraints & Keys	Description
		RESTRICT	
buyer_business_id	UUID	NOT NULL REFERENCES businesses(id) ON DELETE RESTRICT	Foreign Key to buyer businesses ID
buyer_business_name	VARCHAR(200)	NOT NULL	Denormalized buyer company name
buyer_name	VARCHAR(150)	NOT NULL	Buyer representative name
buyer_phone	VARCHAR(30)	NOT NULL	Buyer phone
seller_user_id	UUID	NOT NULL REFERENCES users(id) ON DELETE RESTRICT	Foreign Key to seller users ID
seller_business_id	UUID	NOT NULL REFERENCES businesses(id) ON DELETE RESTRICT	Foreign Key to seller businesses ID
seller_business_name	VARCHAR(200)	NOT NULL	Denormalized seller company name
status	VARCHAR(30)	DEFAULT 'placed'	Status (placed, confirmed, shipped, delivered)
escrow_status	VARCHAR(30)	DEFAULT 'held_in_escrow'	Escrow status (held_in_escrow, released, refunded)
escrow_provider	VARCHAR(50)	DEFAULT 'telebirr'	Escrow provider (telebirr, cbe_birr)
payment_method	VARCHAR(50)	DEFAULT 'telebirr'	Payment method
payment_transaction_ref	VARCHAR(100)	NULLABLE	Financial transaction reference
total_amount	NUMERIC(14, 2)	NOT NULL	Total transaction value
currency	VARCHAR(10)	DEFAULT 'ETB'	Currency code
delivery_address	JSONB	NOT NULL	Complete shipping address JSON
tracking_number	VARCHAR(100)	NULLABLE	Freight tracking code

Column Name	Data Type	Constraints & Keys	Description
freight_carrier	VARCHAR(100)	NULLABLE	Logistics carrier
created_at	TIMESTAMPTZ	DEFAULT CURRENT_TIMESTAMP	Placement timestamp
confirmed_at	TIMESTAMPTZ	NULLABLE	Seller confirmation timestamp
shipped_at	TIMESTAMPTZ	NULLABLE	Dispatch timestamp
delivered_at	TIMESTAMPTZ	NULLABLE	Delivery timestamp
escrow_released_at	TIMESTAMPTZ	NULLABLE	Escrow payout release timestamp

Table: 13. order_items

Individual line items inside a purchase order.

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY DEFAULT gen_random_uuid()	Line item ID
order_id	UUID	NOT NULL REFERENCES orders(id) ON DELETE CASCADE	Foreign Key to orders table
product_id	UUID	NOT NULL REFERENCES products(id) ON DELETE RESTRICT	Foreign Key to products table
product_name	VARCHAR(255)	NOT NULL	Denormalized item title
product_image	TEXT	NULLABLE	Denormalized image thumbnail

Column Name	Data Type	Constraints & Keys	Description
quantity	INTEGER	NOT NULL	Quantity ordered
unit	VARCHAR(50)	NOT NULL	Quantity unit
unit_price	NUMERIC(12, 2)	NOT NULL	Applied unit price
total_price	NUMERIC(14, 2)	NOT NULL	Total line item cost

Table: 14. inquiries

Product-anchored Request for Quotation (RFQ) inquiry threads.

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY DEFAULT gen_random_uuid()	Inquiry thread ID
product_id	UUID	NOT NULL REFERENCES products(id) ON DELETE CASCADE	Foreign Key to products table
product_name	VARCHAR(255)	NOT NULL	Denormalized item title
buyer_user_id	UUID	NOT NULL REFERENCES users(id) ON DELETE CASCADE	Foreign Key to buyer users ID
buyer_business_id	UUID	NOT NULL REFERENCES businesses(id) ON DELETE CASCADE	Foreign Key to buyer businesses ID
buyer_business_name	VARCHAR(200)	NOT NULL	Denormalized buyer company
seller_user_id	UUID	NOT NULL REFERENCES users(id) ON DELETE CASCADE	Foreign Key to seller users ID
seller_business_id	UUID	NOT NULL REFERENCES businesses(id) ON DELETE CASCADE	Foreign Key to seller businesses ID
seller_business_name	VARCHAR(200)	NOT NULL	Denormalized seller company

Column Name	Data Type	Constraints & Keys	Description
topic	VARCHAR(50)	NOT NULL	Inquiry topic (pricing, specifications, sample)
target_quantity	INTEGER	NULLABLE	Target bulk order size
target_price	NUMERIC(12, 2)	NULLABLE	Requested target unit price
status	VARCHAR(30)	DEFAULT 'pending_reply'	Thread status (pending_reply, replied, closed)
created_at	TIMESTAMPTZ	DEFAULT CURRENT_TIMESTAMP	Thread start timestamp

Table: 15. inquiry_messages

Messages exchanged within an RFQ inquiry thread.

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY DEFAULT gen_random_uuid()	Message ID
inquiry_id	UUID	NOT NULL REFERENCES inquiries(id) ON DELETE CASCADE	Foreign Key to inquiries table
sender_user_id	UUID	NOT NULL REFERENCES users(id) ON DELETE CASCADE	Foreign Key to sender users ID
sender_name	VARCHAR(150)	NOT NULL	Sender full name
sender_business	VARCHAR(200)	NOT NULL	Sender business name

Column Name	Data Type	Constraints & Keys	Description
role_in_thread	VARCHAR(20)	NOT NULL	Sender role (buyer / seller)
text	TEXT	NOT NULL	Message text content
created_at	TIMESTAMPTZ	DEFAULT CURRENT_TIMESTAMP	Message timestamp

Table: 16. notifications

User in-app notifications and system alerts.

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY DEFAULT gen_random_uuid()	Notification ID
user_id	UUID	NOT NULL REFERENCES users(id) ON DELETE CASCADE	Foreign Key to recipient users ID
title	VARCHAR(200)	NOT NULL	Alert headline
message	TEXT	NOT NULL	Alert body text
type	VARCHAR(50)	NOT NULL	Type (order_status, inquiry_reply, escrow_update)
read	BOOLEAN	DEFAULT FALSE	Read status flag
created_at	TIMESTAMPTZ	DEFAULT CURRENT_TIMESTAMP	Notification timestamp

Table: 17. sms_logs

Audit log of outbound SMS verification OTPs and escrow alerts.

Column Name	Data Type	Constraints & Keys	Description
id	UUID	PRIMARY KEY DEFAULT gen_random_uuid()	Log entry ID
phone	VARCHAR(30)	NOT NULL	Recipient phone number
message	TEXT	NOT NULL	SMS content

Column Name	Data Type	Constraints & Keys	Description
sent_at	TIMESTAMPTZ	DEFAULT CURRENT_TIMESTAMP	Outbound dispatch timestamp